RUDRA PRATAP SINGH

AI and Full-Stack Developer | Agentic AI, RAG, LLM Security and Evaluation

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Portfolio: rudraksh01-tech.github.io

SUMMARY

Computer Science undergraduate (VTU, CGPA 8.0) building Generative AI applications with LLMs. Shipped three live, evaluated projects: an AI-agent security gateway, a multilingual RAG assistant and an agentic AI assistant. Founding Software Engineer Intern at FERMOR; seeking AI and full-stack internships.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering (CSE) 2024 - 2028 (Expected) | 5th semester
Cambridge Institute of Technology, North Campus, Bengaluru
Visvesvaraya Technological University (VTU) Current CGPA: 8.0 / 10

Class XII (CBSE) | Physics, Chemistry and Mathematics (PCM) 74%

Class X (CBSE) 74%

EXPERIENCE

Founding Software Engineer Intern, FERMOR Sep 2026 - Oct 2026
Bengaluru, Karnataka (Hybrid)

- Redesigned key website pages end to end with a cleaner frosted-glass UI/UX, and optimised structure, headings, metadata and content for SEO.
- Drove about 30x the usual daily traffic on launch day (a month's worth in one day) through the redesign and SEO work; used analytics to measure the impact and guide next steps.

PROJECTS

Dwarpal | Security Gateway for AI Agents (MCP) rudraksh01-tech.github.io/dwarpal
Python, LangGraph, MCP, FastAPI, PostgreSQL (pgvector), Redis, Docker, Kubernetes, Helm, React | github.com/rudraksh01-tech/dwarpal

- Built an open-source gateway between AI agents and their MCP tool servers that blocks tool poisoning, rug pulls and cross-server data leaks in English, Hindi, Hinglish and Kannada; runs as an MCP server for any client.
- Designed a LangGraph policy graph: SHA-256 tool pinning, taint tracking of private data across servers, multilingual rules, Prompt Guard 2 and an LLM policy judge, with cached verdicts and human approval.
- Measured over repeated runs: the unprotected agent leaked user data in 5 of 25 attack runs, 0 of 25 with Dwarpal, no false blocks; Prompt Guard 2 scored the polite attack 0.001 (safe).
- Stored tool pins and audit logs in PostgreSQL with pgvector, cached in Redis; CI installs and tests the Helm chart on Kubernetes (kind).

AyurIP-Sahayak | Multilingual RAG Assistant ayur-ip-sahayak.vercel.app
Python, FastAPI, React, Google Gemini API, NumPy, Docker, GitHub Actions | github.com/rudraksh01-tech/AyurIP-Sahayak

- Built a RAG assistant that answers questions on Ayurveda, traditional knowledge and Indian IP law in English, Hindi, Kannada and Hinglish, citing the exact source page for every claim.
- Wrote retrieval from scratch (section-aware chunking, Gemini embeddings, vector search, BM25 + RRF hybrid); on 38 labelled questions it scored Hit@1 0.95 and MRR 0.97, against 0.47 and 0.60 for a BM25 baseline.
- Added an LLM-as-a-judge evaluation from a different model family: 0.99 faithfulness (166 of 167 claims), 0.98 citation precision and 15 of 15 planted errors caught; Docker, 47 tests and CI.

NestFinder | Agentic AI Assistant for Flat Search nestfinder-woad.vercel.app
Node.js, Express.js, React, MongoDB, Groq API, Langfuse, OpenTelemetry | github.com/rudraksh01-tech/nestfinder

- Developed a tool-calling agent from scratch (no framework): 5 tools, streaming, a repair loop for rejected calls and fallback to a second model on rate limits.
- Wrote a 20-case agent evaluation covering Hindi, Hinglish and prompt-injection inputs; fixed a schema bug it exposed, raising tool-choice accuracy from 18/20 to 20/20.
- Traced every LLM call with OpenTelemetry and Langfuse (phone numbers masked); JWT auth, rate limiting, CI; deployed on Vercel and Render.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

AI / LLM: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangGraph, Model Context Protocol (MCP), Tool Calling, AI Security (Prompt Injection, Guardrails), Embeddings, Vector Search, Hybrid Search (BM25, RRF), LLM Evaluation (LLM-as-a-Judge), Prompt Engineering, Gemini API, Groq API

Backend: FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, pgvector, Redis, MongoDB, JWT Authentication

Frontend: React, Vite, Tailwind CSS, HTML, CSS

DevOps & Tools: Docker, Kubernetes, Helm, GitHub Actions (CI/CD), Git, Linux, pytest, Vercel, Render, Langfuse, OpenTelemetry

Core CS: Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks